IBM SkillsBuild – AICTE-2026

PROJECT PROBLEM STATEMENT

Problem Statement ID: 29

Problem Statement Title: Street Vendor Digitalization Agent

Domain: Small Business Digitalization · Generative AI / Agentic AI

Problem Statement:

Street vendors, hawkers and small local businesses in India mostly work offline, with no online shop profile or catalogue, little visibility on search or social media, and limited knowledge of UPI payments and government schemes. Many prefer Hindi or Marathi and have limited technical experience, so generic chatbots are hard to use. The problem is to build a Street Vendor Digitalization Agent that uses IBM Granite to help vendors set up a digital presence and promote their businesses through a safe, easy-to-use AI assistant, while keeping the vendor in control and never handling payment secrets. The project addresses this problem.

Project Name: DukaanSaathi AI

Objectives / Requirements:

- Guided onboarding for shop name, business type, products, prices, opening hours, preferred language and optional service area.
- Generate an editable digital shop profile and a product catalogue the vendor can add, edit and delete.
- Use IBM Granite to write product descriptions, WhatsApp promotions, festival offers and social media captions in English, Hindi and Marathi.
- Give practical digital marketing, customer engagement and local SEO suggestions, labelled as suggestions, not guarantees.
- Guide vendors step by step on UPI and payment QR through legitimate providers, never collecting UPI PINs, OTPs or passwords.
- Show government scheme information from verified official sources with links and eligibility caveats, never inventing benefits.
- Follow a controlled workflow (understand request, gather context, generate, validate, vendor approval) and never submit applications, send promotions or make payments automatically.
- Allow editing, previewing and downloading of the profile and content, with a dashboard, demo mode, validation and clear error messages.
- Use IBM Granite through IBM watsonx.ai (IBM Cloud Lite), with no hardcoded API keys.

Expected Outcome:

A working, responsive web application (FastAPI backend, HTML/CSS/JavaScript frontend) that lets a vendor create a digital shop profile and catalogue, generate and approve marketing content with IBM Granite in their language, learn safe UPI use, and find official scheme information. The deliverables are the complete source code, setup instructions and .env.example, API documentation and tests, an architecture and data-flow description, an explanation of the IBM Granite and IBM Cloud Lite integration, a Responsible AI section, and a 2–3 minute demonstration script with a presentation outline.

Source:

IBM SkillsBuild – AICTE-2026 problem statements (Problem Statement #29)